

Strong Duality Beyond Affine Costs in Distorted Optimal Transport

Universal Distortions and a Sharp Monotone Cost Class

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Abstract

Liu, Wang, Wang, and Zhuang asked whether their strong-duality theorem for distorted optimal transport extends beyond affine costs. We give substantial partial progress, but not a complete solution of the open-ended classification problem. First, every bounded separable cost satisfies strong duality for every convex distortion. Second, we solve the following universal subproblem: we characterize the distortions for which the same dual formula is valid for every bounded continuous cost and all marginals. They are exactly

$$h_\lambda(t) = \frac{(t + \lambda - 1)_+}{\lambda}, \quad 0 < \lambda \leq 1,$$

together with the limiting distortion $h_0 = \mathbf{1}_{\{1\}}$. The proof reduces the primal problem to a partial-transport problem and then uses classical Kantorovich duality and Sion's minimax theorem. Third, for an arbitrary convex distortion, strong duality holds for every bounded continuous componentwise monotone submodular cost. For continuous, strictly increasing, convex, nonaffine distortions, submodularity is also necessary within the componentwise-monotone class if duality is required for all marginals. Thus the paper completely answers two natural restricted classification questions while leaving the classification of all pairs (h, c) open.

Status of the open problem. The results below are *partial progress*, not a complete solution of the question in [2, Section 9]. They do, however, give: (i) a complete characterization of the distortions for which no restriction on a bounded continuous cost is needed; and (ii) a sharp characterization of the costs within the bounded continuous componentwise-monotone class for a broad class of strictly increasing convex distortions.

1 Introduction

Let μ and ν be probability measures on $\mathbb{R}$, let $c : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a transport cost, and let $h : [0, 1] \rightarrow [0, 1]$ be a distortion. Distorted optimal transport replaces ordinary expectation by the Choquet integral $\mathbb{E}_h$ and considers

$$\inf_{\pi \in \Pi(\mu, \nu)} \mathbb{E}_h^\pi[c].$$

For convex h , Liu, Wang, Wang, and Zhuang established weak duality for the natural Kantorovich-type dual and strong duality for affine costs [2]. They asked whether strong duality persists more generally.

The question is deliberately broad: it asks for conditions on both h and c . Consequently, a few examples beyond affine costs do not amount to a full solution, and neither does a characterization obtained under a strong extra quantifier. The purpose of this note is to isolate and solve two substantial subproblems.

The first subproblem places the strongest possible quantifier on the cost: for which distortions does duality hold for *every* bounded continuous c ? The answer is the lower-tail-average family h_λ , including ordinary expectation at $\lambda = 1$ and essential infimum at $\lambda = 0$. Necessity already follows from a two-point “AND” cost. Sufficiency follows from a robust representation of the lower-tail average, an exact partial-transport reduction, classical Kantorovich duality, and a minimax interchange.

The second subproblem fixes an arbitrary convex distortion and asks for a broad cost class. The source paper proves that the comonotone coupling minimizes the distorted cost when c is componentwise monotone and submodular. We combine that primal theorem with a bounded monotone splitting-potential construction to obtain strong duality and dual attainment. A two-point obstruction then shows that, for continuous strictly increasing convex nonaffine distortions, submodularity is necessary within the monotone class.

The remaining sections make these statements precise. All random variables and costs used inside distorted expectations are bounded, so no integrability or extended-value convention is needed.

2 Setup and preliminary facts

A *distortion* is an increasing map $h : [0, 1] \rightarrow [0, 1]$ satisfying $h(0) = 0$ and $h(1) = 1$. For a bounded random variable Z , its distorted expectation is

$$\mathbb{E}_h[Z] = \int_0^\infty h(\mathbb{P}(Z > t)) dt + \int_{-\infty}^0 (h(\mathbb{P}(Z > t)) - 1) dt. \quad (1)$$

We use a superscript to indicate the underlying law when useful. Thus $\mathbb{E}_h^\pi[c]$ is the distorted expectation of $c(X, Y)$ when (X, Y) has law π .

For probability measures $\mu, \nu \in \mathcal{P}(\mathbb{R})$, let $\Pi(\mu, \nu)$ be the set of their couplings. For a bounded Borel cost c , define

$$P_h(c; \mu, \nu) := \inf_{\pi \in \Pi(\mu, \nu)} \mathbb{E}_h^\pi[c], \quad (2)$$

$$D_h(c; \mu, \nu) := \sup_{(\varphi, \psi) \in \mathcal{S}_c^b} \{ \mathbb{E}_h^\mu[\varphi] + \mathbb{E}_h^\nu[\psi] \}, \quad (3)$$

where

$$\mathcal{S}_c^b := \{ (\varphi, \psi) \in \mathcal{B}_b(\mathbb{R})^2 : \varphi(x) + \psi(y) \leq c(x, y) \text{ for all } x, y \}.$$

Here $\mathcal{B}_b(\mathbb{R})$ denotes the bounded Borel functions. Equality of (2) and (3) is the strong-duality identity studied below. Since the dual class is bounded, all terms are finite. If one enlarges the admissible class to functions for which weak duality remains valid, any identity proved using bounded potentials remains an identity for the enlarged dual as well.

Two random variables U, V are *comonotone* if there is a random variable W and two increasing functions f, g such that $U = f(W)$ and $V = g(W)$.

Lemma 2.1 (Basic Choquet properties). *For every distortion h , the functional $\mathbb{E}_h$ is monotone, cash invariant, positively homogeneous, and comonotonically additive. In particular, if U, V are comonotone, then*

$$\mathbb{E}_h[U + V] = \mathbb{E}_h[U] + \mathbb{E}_h[V].$$

If h is convex, then $\mathbb{E}_h$ is superadditive:

$$\mathbb{E}_h[U + V] \geq \mathbb{E}_h[U] + \mathbb{E}_h[V].$$

Consequently, for convex h ,

$$D_h(c; \mu, \nu) \leq P_h(c; \mu, \nu). \quad (4)$$

Proof. The first assertions are standard properties of the Choquet integral; see [1]. If h is convex, then the capacity $A \mapsto h(\mathbb{P}(A))$ is supermodular, and its Choquet integral is superadditive. For $(\varphi, \psi) \in \mathcal{S}_c^b$ and any coupling (X, Y) , monotonicity and superadditivity give

$$\mathbb{E}_h[c(X, Y)] \geq \mathbb{E}_h[\varphi(X) + \psi(Y)] \geq \mathbb{E}_h[\varphi(X)] + \mathbb{E}_h[\psi(Y)].$$

Taking first the infimum over couplings and then the supremum over feasible potentials proves (4). $\square$

The next standard lifting fact will be useful twice.

Lemma 2.2 (Lifting couplings through measurable maps). *Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be Borel, let $\alpha = f_{\#}\mu$ and $\beta = g_{\#}\nu$, and let $\rho \in \Pi(\alpha, \beta)$. There exists $\pi \in \Pi(\mu, \nu)$ such that $(f, g)_{\#}\pi = \rho$.*

Proof. Because $\mathbb{R}$ is a standard Borel space, disintegrations exist:

$$\mu(dx) = K_z(dx) \alpha(dz), \quad \nu(dy) = L_w(dy) \beta(dw),$$

where K_z is concentrated on $f^{-1}(\{z\})$ for α -almost every z , and L_w is concentrated on $g^{-1}(\{w\})$ for β -almost every w . Define

$$\pi(dx, dy) := \int_{\mathbb{R}^2} K_z(dx) L_w(dy) \rho(dz, dw).$$

Its marginals are μ and ν , and its image under (f, g) is ρ . $\square$

3 A preliminary extension: separable costs

The affine theorem extends immediately to arbitrary bounded separable costs. The only small point is that a comonotone coupling of the transformed variables must be lifted to the original spaces.

Proposition 3.1 (Separable costs). *Let h be convex, let $f, g \in \mathcal{B}_b(\mathbb{R})$, and set $c(x, y) = f(x) + g(y)$. Then for every $\mu, \nu \in \mathcal{P}(\mathbb{R})$,*

$$P_h(c; \mu, \nu) = D_h(c; \mu, \nu) = \mathbb{E}_h^\mu[f] + \mathbb{E}_h^\nu[g]. \quad (5)$$

Proof. The pair (f, g) is feasible, so weak duality gives

$$\mathbb{E}_h^\mu[f] + \mathbb{E}_h^\nu[g] \leq D_h(c; \mu, \nu) \leq P_h(c; \mu, \nu).$$

Let $\alpha = f_{\#}\mu$ and $\beta = g_{\#}\nu$. Couple $U \sim \alpha$ and $V \sim \beta$ comonotonically, and use Lemma 2.2 to obtain a coupling (X, Y) of (μ, ν) with $(f(X), g(Y)) = (U, V)$ in law. By comonotonic additivity,

$$\mathbb{E}_h[c(X, Y)] = \mathbb{E}_h[U + V] = \mathbb{E}_h[U] + \mathbb{E}_h[V] = \mathbb{E}_h^\mu[f] + \mathbb{E}_h^\nu[g].$$

This proves the reverse inequality for the primal. $\square$

4 A universal-cost characterization

For $0 < \lambda \leq 1$, define

$$h_\lambda(t) := \frac{(t + \lambda - 1)_+}{\lambda}, \quad 0 \leq t \leq 1, \quad (6)$$

and define the limiting distortion

$$h_0(t) := \mathbf{1}_{\{1\}}(t). \quad (7)$$

For $\lambda > 0$, $\mathbb{E}_{h_\lambda}$ is the average of the lowest λ -fraction of the quantiles. The endpoint $h_1(t) = t$ gives ordinary expectation, and h_0 gives essential infimum.

Theorem 4.1 (Universal-cost classification). *For a distortion h , the following statements are equivalent.*

(i) *For every $\mu, \nu \in \mathcal{P}(\mathbb{R})$ and every bounded continuous $c : \mathbb{R}^2 \rightarrow \mathbb{R}$,*

$$\mathbf{P}_h(c; \mu, \nu) = \mathbf{D}_h(c; \mu, \nu).$$

(ii) *There exists $\lambda \in [0, 1]$ such that $h = h_\lambda$, with h_0 interpreted as in (7).*

4.1 Necessity: a two-point test

Let $T(a, b) = (a + b - 1)_+$ be the Łukasiewicz operation.

Lemma 4.2 (The AND-cost test). *Choose a continuous $s : \mathbb{R} \rightarrow [0, 1]$ satisfying $s(0) = 0$ and $s(1) = 1$, and set $c(x, y) = s(x)s(y)$. Let μ and ν be supported on $\{0, 1\}$ with $\mu\{1\} = a$ and $\nu\{1\} = b$. Then*

$$\mathbf{P}_h(c; \mu, \nu) = h(T(a, b)), \quad (8)$$

$$\mathbf{D}_h(c; \mu, \nu) = T(h(a), h(b)). \quad (9)$$

Proof. For a coupling of the two Bernoulli marginals, the mass r of the cell $(1, 1)$ ranges over $[T(a, b), \min(a, b)]$. The cost is Bernoulli with success probability r , and hence has distorted expectation $h(r)$. Since h is increasing, (8) follows.

For the dual upper bound, restrict any globally feasible pair to $\{0, 1\}$ and write its values as u_0, u_1, v_0, v_1 . The four constraints are

$$u_0 + v_0 \leq 0, \quad u_0 + v_1 \leq 0, \quad u_1 + v_0 \leq 0, \quad u_1 + v_1 \leq 1.$$

For this finite matrix, replacing u by its c -transform and then v by its c -transform raises the potentials coordinatewise, preserves feasibility, and produces $u_0 \leq u_1$ and $v_0 \leq v_1$. By cash invariance, normalize $u_0 = 0$ and write $A = u_1 \geq 0$. The largest feasible values of v are

$$(v_0, v_1) = \begin{cases} (-A, 0), & 0 \leq A \leq 1, \\ (-A, 1 - A), & A \geq 1. \end{cases}$$

The resulting objective is

$$\begin{cases} A(h(a) + h(b) - 1), & 0 \leq A \leq 1, \\ h(b) - A(1 - h(a)), & A \geq 1. \end{cases}$$

Its maximum is $(h(a) + h(b) - 1)_+$.

This finite upper bound is attained by globally feasible potentials. If $h(a) + h(b) \leq 1$, use $(\varphi, \psi) = (0, 0)$. Otherwise, use $\varphi = s$ and $\psi = s - 1$. Feasibility follows from

$$s(x) + s(y) - 1 \leq s(x)s(y),$$

and the objective is $h(a) + h(b) - 1$. This proves (9). $\square$

If universal strong duality holds, Lemma 4.2 implies

$$h(T(a, b)) = T(h(a), h(b)) \quad (a, b \in [0, 1]). \quad (10)$$

Lemma 4.3 (Endomorphisms of the Łukasiewicz operation). *An increasing $h : [0, 1] \rightarrow [0, 1]$ with $h(0) = 0$ and $h(1) = 1$ satisfies (10) if and only if $h = h_\lambda$ for some $\lambda \in [0, 1]$.*

Proof. Set $k(x) = 1 - h(1 - x)$. Equation (10) is equivalent to

$$k(\min\{x + y, 1\}) = \min\{k(x) + k(y), 1\}, \quad x, y \in [0, 1]. \quad (11)$$

If $k(x) = 1$ for every $x > 0$, then $h = h_0$.

Assume otherwise and define

$$\lambda := \sup\{x \in [0, 1] : k(x) < 1\} > 0.$$

Monotonicity implies $k(x) < 1$ for $x < \lambda$ and $k(x) = 1$ for $x > \lambda$. If $x + y < \lambda$, then the left side of (11) is strictly below 1, so no truncation occurs and

$$k(x + y) = k(x) + k(y). \quad (12)$$

We briefly justify the standard local-Cauchy step. Fix $a < \lambda$. On $[0, a]$, equation (12) implies $k(ra) = rk(a)$ for rational $r \in [0, 1]$; monotonicity then extends this identity to all real $r \in [0, 1]$. Comparing two choices of a on their common interval shows that the slope is independent of a . Hence

$$k(x) = cx \quad (0 \leq x < \lambda)$$

for some $c \geq 0$.

Since $k(x) < 1$ for $x < \lambda$, letting $x \uparrow \lambda$ gives $c\lambda \leq 1$. If $\lambda < 1$, choose $x, y < \lambda$ with $x + y > \lambda$ and let $x + y \downarrow \lambda$. Equation (11) gives $\min\{c(x + y), 1\} = 1$, and hence $c\lambda \geq 1$. If $\lambda = 1$, choose $x, y < 1$ with $x + y = 1$ and obtain the same conclusion. Thus $c\lambda = 1$. Taking $x = y = \lambda/2$ in (11) also gives $k(\lambda) = 1$. Therefore

$$k(x) = \min\left\{\frac{x}{\lambda}, 1\right\}.$$

Transforming back yields $h = h_\lambda$. Direct substitution verifies that every h_λ satisfies (10). $\square$

Lemmas 4.2 and 4.3 prove necessity in Theorem 4.1. No regularity or convexity assumption on h was used. The conclusion forces convexity, but it does *not* force continuity: h_0 is discontinuous at 1.

4.2 Sufficiency for $0 < \lambda \leq 1$

For $0 < \lambda \leq 1$ and $\mu \in \mathcal{P}(\mathbb{R})$, define

$$\mathcal{A}_\lambda(\mu) := \{\alpha \in \mathcal{P}(\mathbb{R}) : \lambda\alpha \leq \mu\}, \quad (13)$$

where the inequality is domination of finite measures.

For a bounded random variable with law μ , let $Q_f(u) = \inf\{t : \mu(f \leq t) \geq u\}$ be the lower quantile of f .

Lemma 4.4 (Lower-tail representation). *For every bounded Borel f ,*

$$\mathbb{E}_{h_\lambda}^\mu[f] = \frac{1}{\lambda} \int_0^\lambda Q_f(u) \, du = \inf_{\alpha \in \mathcal{A}_\lambda(\mu)} \int f \, d\alpha. \quad (14)$$

Moreover, $\mathcal{A}_\lambda(\mu)$ is convex and weakly compact.

Proof. The first equality follows directly from the quantile representation of the Choquet integral, since h_λ has slope $1/\lambda$ on $[1 - \lambda, 1]$ and is zero below $1 - \lambda$.

Write $\alpha = \xi\mu$. The constraint $\lambda\alpha \leq \mu$ is equivalent to

$$0 \leq \xi \leq \frac{1}{\lambda}, \quad \int \xi \, d\mu = 1.$$

Let $q = Q_f(\lambda)$. Choose $\theta \in [0, 1]$ so that

$$\xi_* := \frac{1}{\lambda} \mathbf{1}_{\{f < q\}} + \frac{\theta}{\lambda} \mathbf{1}_{\{f = q\}}$$

has integral one. For any feasible ξ , the sign of $f - q$ gives

$$\int (f - q) \xi \, d\mu \geq \frac{1}{\lambda} \int_{\{f < q\}} (f - q) \, d\mu.$$

Equality is attained by ξ_* . The resulting value is exactly $\lambda^{-1} \int_0^\lambda Q_f(u) \, du$, proving the variational representation.

Convexity of $\mathcal{A}_\lambda(\mu)$ is immediate. It is uniformly tight because $\alpha(K^c) \leq \mu(K^c)/\lambda$. It is weakly closed: if $\alpha_n \Rightarrow \alpha$ and $\lambda\alpha_n \leq \mu$, then for every nonnegative $g \in C_b(\mathbb{R})$,

$$\lambda \int g \, d\alpha = \lim_n \lambda \int g \, d\alpha_n \leq \int g \, d\mu,$$

which implies $\lambda\alpha \leq \mu$. Prokhorov's theorem now gives weak compactness. $\square$

Applying Lemma 4.4 on the product space gives, for each $\pi \in \Pi(\mu, \nu)$,

$$\mathbb{E}_{h_\lambda}^\pi[c] = \inf \left\{ \int c \, d\gamma : \gamma \in \mathcal{P}(\mathbb{R}^2), \lambda\gamma \leq \pi \right\}.$$

Eliminating π yields an exact partial-transport formulation.

Lemma 4.5 (Partial-transport reduction). *For bounded Borel c ,*

$$P_{h_\lambda}(c; \mu, \nu) = \inf_{\substack{\gamma \in \mathcal{P}(\mathbb{R}^2) \\ \lambda\gamma_1 \leq \mu, \lambda\gamma_2 \leq \nu}} \int c \, d\gamma. \quad (15)$$

Proof. If $\lambda\gamma \leq \pi$ for some $\pi \in \Pi(\mu, \nu)$, then the two marginal domination constraints in (15) follow. Conversely, suppose those constraints hold. The residual measures $\mu - \lambda\gamma_1$ and $\nu - \lambda\gamma_2$ both have mass $1 - \lambda$. Couple them by any finite measure η of mass $1 - \lambda$ (with $\eta = 0$ when $\lambda = 1$) and set $\pi = \lambda\gamma + \eta$. Then $\pi \in \Pi(\mu, \nu)$ and $\lambda\gamma \leq \pi$. $\square$

Proof of sufficiency in Theorem 4.1 for $0 < \lambda \leq 1$. Let

$$\mathcal{K}_c := \{(\varphi, \psi) \in C_b(\mathbb{R})^2 : \varphi(x) + \psi(y) \leq c(x, y) \ \forall x, y\}.$$

By Lemma 4.5 and classical Kantorovich duality for bounded continuous costs on Polish spaces [4],

$$\begin{aligned} P_{h_\lambda}(c; \mu, \nu) &= \inf_{\substack{\alpha \in \mathcal{A}_\lambda(\mu) \\ \beta \in \mathcal{A}_\lambda(\nu)}} \inf_{\gamma \in \Pi(\alpha, \beta)} \int c \, d\gamma \\ &= \inf_{\alpha, \beta} \sup_{(\varphi, \psi) \in \mathcal{K}_c} \left(\int \varphi \, d\alpha + \int \psi \, d\beta \right). \end{aligned} \quad (16)$$

Equip $\mathcal{A}_\lambda(\mu) \times \mathcal{A}_\lambda(\nu)$ with the product weak topology and $C_b(\mathbb{R})^2$ with the product sup norm. The former set is compact and convex by Lemma 4.4, and $\mathcal{K}_c$ is convex. The payoff in (16) is finite, continuous, and affine in each variable. Sion's minimax theorem [3] therefore yields

$$\begin{aligned} P_{h_\lambda}(c; \mu, \nu) &= \sup_{(\varphi, \psi) \in \mathcal{K}_c} \left\{ \inf_{\alpha \in \mathcal{A}_\lambda(\mu)} \int \varphi \, d\alpha + \inf_{\beta \in \mathcal{A}_\lambda(\nu)} \int \psi \, d\beta \right\} \\ &= \sup_{(\varphi, \psi) \in \mathcal{K}_c} \left\{ \mathbb{E}_{h_\lambda}^\mu[\varphi] + \mathbb{E}_{h_\lambda}^\nu[\psi] \right\} \\ &\leq D_{h_\lambda}(c; \mu, \nu). \end{aligned}$$

The distortion h_λ is convex, so weak duality gives the reverse inequality. Hence strong duality holds. $\square$

4.3 The endpoint $\lambda = 0$

Proof of sufficiency in Theorem 4.1 for $\lambda = 0$. For h_0 , formula (1) gives $\mathbb{E}_{h_0}[Z] = \text{ess inf } Z$. Let $S = \text{supp } \mu$, $T = \text{supp } \nu$, and

$$m := \inf_{(x, y) \in S \times T} c(x, y).$$

Every coupling is concentrated on $S \times T$, so its essential infimum is at least m . Conversely, for every $\varepsilon > 0$, choose $(x_0, y_0) \in S \times T$ with $c(x_0, y_0) < m + \varepsilon/2$. By continuity, there are neighborhoods $U \ni x_0$ and $V \ni y_0$ on which $c < m + \varepsilon$. Since $x_0 \in S$ and $y_0 \in T$, the product coupling $\mu \otimes \nu$ gives positive mass to $U \times V$, and hence its essential infimum is at most $m + \varepsilon$. Thus

$$P_{h_0}(c; \mu, \nu) = m. \quad (17)$$

Let $L = \inf_{\mathbb{R}^2} c$ and define

$$\varphi(x) = \begin{cases} m, & x \in S, \\ L, & x \notin S, \end{cases} \quad \psi(y) = \begin{cases} 0, & y \in T, \\ L - m, & y \notin T. \end{cases}$$

The sets S, T are closed, so the potentials are bounded Borel. They are feasible. Indeed, on $S \times T$ their sum is $m \leq c$; if exactly one coordinate lies outside its support, their sum is $L \leq c$; and if both lie outside, their sum is $2L - m \leq L \leq c$. Moreover,

$$\text{ess inf}_\mu \varphi = m, \quad \text{ess inf}_\nu \psi = 0.$$

Hence the dual value is at least m , while weak duality for essential infimum gives the reverse inequality. Together with (17), this proves strong duality and completes Theorem 4.1. $\square$

Example 4.6 (A nonlinear universal distortion). *For $h(t) = h_{1/2}(t) = (2t - 1)_+$, strong duality holds for every bounded continuous cost, including $c(x, y) = \min\{1, |x - y|\}$, which is neither affine nor componentwise monotone.*

5 Arbitrary convex distortions: monotone submodular costs

Definition 5.1. A function $c : \mathbb{R}^2 \rightarrow \mathbb{R}$ is *submodular* if

$$c(x_0, y_0) + c(x_1, y_1) \leq c(x_0, y_1) + c(x_1, y_0) \quad (18)$$

whenever $x_0 \leq x_1$ and $y_0 \leq y_1$. It is *componentwise monotone* if it is nondecreasing in both coordinates or nonincreasing in both coordinates.

Let π^+ be the comonotone coupling of μ and ν , namely the law of

$$(F_\mu^{-1}(U), F_\nu^{-1}(U)), \quad U \sim \text{Unif}(0, 1).$$

The following primal theorem is due to Liu, Wang, Wang, and Zhuang.

Theorem 5.2 (Comonotone optimality [2]). *If h is convex and c is componentwise monotone and submodular, then π^+ minimizes $\mathbb{E}_h^\pi[c]$ over $\pi \in \Pi(\mu, \nu)$.*

We next develop the dual potentials needed to turn this primal statement into strong duality. A set $\Gamma \subseteq \mathbb{R}^2$ is c -cyclically monotone if for every finite family $(x_i, y_i) \in \Gamma$ and every permutation σ ,

$$\sum_i c(x_i, y_i) \leq \sum_i c(x_i, y_{\sigma(i)}).$$

Lemma 5.3 (Bounded monotone splitting potentials). *Let $c \in C_b(\mathbb{R}^2)$ be componentwise nondecreasing, and let $\Gamma \subseteq \mathbb{R}^2$ be a nonempty c -cyclically monotone set. Then there exist bounded, upper semicontinuous, nondecreasing functions $\varphi, \psi : \mathbb{R} \rightarrow \mathbb{R}$ such that*

$$\varphi(x) + \psi(y) \leq c(x, y) \quad \text{for all } x, y, \quad \varphi(x) + \psi(y) = c(x, y) \quad \text{on } \Gamma. \quad (19)$$

For componentwise nonincreasing c , the potentials may be chosen nonincreasing.

Proof. We treat the nondecreasing case. By the classical two-marginal splitting theorem for finite-valued costs, a c -cyclically monotone set is a c -splitting set; see [4, Chapter 5]. Thus, with $A = \text{proj}_1 \Gamma$ and $B = \text{proj}_2 \Gamma$, there are real-valued functions φ_0 on A and ψ_0 on B such that

$$\varphi_0(x) + \psi_0(y) \leq c(x, y) \quad ((x, y) \in A \times B),$$

with equality on Γ .

These initial potentials have bounded oscillation. Indeed, if $(x, y), (x', y') \in \Gamma$, equality at those two points and feasibility at the crossed points give

$$c(x, y) - c(x, y') \leq \psi_0(y) - \psi_0(y') \leq c(x', y) - c(x', y').$$

Hence $|\psi_0(y) - \psi_0(y')| \leq \text{osc}(c)$, and the analogous estimate holds for φ_0 . After a common additive normalization, both functions are bounded.

Define successive c -transforms by

$$\begin{aligned}\varphi(x) &:= \inf_{y \in B} (c(x, y) - \psi_0(y)), \\ \psi(y) &:= \inf_{x \in \mathbb{R}} (c(x, y) - \varphi(x)).\end{aligned}$$

Both functions are finite and bounded. The first transform agrees with φ_0 on A : feasibility gives $\varphi \geq \varphi_0$ there, while a point of Γ over each $x \in A$ gives the reverse inequality. Moreover, for every $y \in B$ the first definition gives $\varphi(x) + \psi_0(y) \leq c(x, y)$ for all x . Therefore the second transform satisfies $\psi \geq \psi_0$ on B , while equality at a point of Γ over y gives $\psi \leq \psi_0$. Thus equality in (19) is preserved on Γ , and global feasibility follows from the definition of ψ .

Because c is nondecreasing in each coordinate, infima in the displayed formulas preserve non-decreasingness. Each potential is an infimum of a family of continuous one-variable functions, and is therefore upper semicontinuous. The nonincreasing case is identical with all monotonicities reversed. $\square$

Theorem 5.4 (Strong duality for monotone submodular costs). *Let h be convex, and let $c : \mathbb{R}^2 \rightarrow \mathbb{R}$ be bounded, continuous, componentwise monotone, and submodular. Then for every $\mu, \nu \in \mathcal{P}(\mathbb{R})$,*

$$P_h(c; \mu, \nu) = D_h(c; \mu, \nu). \quad (20)$$

Moreover, the comonotone coupling attains the primal. The dual admits a bounded maximizing pair whose components are monotone in the same direction as c .

Proof. We treat the nondecreasing case. Let Γ be the monotone image of $u \mapsto (F_\mu^{-1}(u), F_\nu^{-1}(u))$; it carries π^+ . The set Γ is c -cyclically monotone. To see this, order a finite family from Γ so that both coordinates are nondecreasing. If a permutation has an inversion, swapping the two inverted y -coordinates weakly decreases the permuted cost by submodularity. Repeating the operation sorts the permutation and proves the cyclic inequality.

Apply Lemma 5.3 to obtain bounded nondecreasing potentials (φ, ψ) satisfying (19). Let (X, Y) have law π^+ . By Theorem 5.2,

$$P_h(c; \mu, \nu) = \mathbb{E}_h[c(X, Y)].$$

Since $c(X, Y) = \varphi(X) + \psi(Y)$ almost surely on Γ , and since $\varphi(X)$ and $\psi(Y)$ are comonotone,

$$\begin{aligned}P_h(c; \mu, \nu) &= \mathbb{E}_h[\varphi(X) + \psi(Y)] \\ &= \mathbb{E}_h^\mu[\varphi] + \mathbb{E}_h^\nu[\psi] \\ &\leq D_h(c; \mu, \nu).\end{aligned}$$

Weak duality gives the reverse inequality. The nonincreasing case is the same: both transformed variables are decreasing functions of the common quantile parameter and hence are still comonotone. $\square$

Example 5.5 (A genuinely nonseparable class). *The cost*

$$c(x, y) = 1 - \exp\{-e^x - e^y\}$$

is bounded and strictly increasing, and

$$\frac{\partial^2 c}{\partial x \partial y} = -e^{x+y} \exp\{-e^x - e^y\} < 0.$$

It is therefore strictly submodular and satisfies strong duality for every convex distortion and every pair of marginals.

6 Sharpness within the monotone class

We now show that the preceding cost condition is necessary for genuinely nonlinear strictly increasing convex distortions.

Lemma 6.1 (A strict convexity increment). *Let h be continuous, convex, and nonaffine. Then there exist $p, q > 0$ with $p + q < 1$ such that*

$$\Delta_h(p, q) := 1 + h(1 - p - q) - h(1 - p) - h(1 - q) > 0. \quad (21)$$

Proof. For fixed $q > 0$, convexity implies that $x \mapsto h(x + q) - h(x)$ is nondecreasing on $[0, 1 - q]$. Therefore $\Delta_h(p, q) \geq 0$ whenever $p, q > 0$ and $p + q < 1$.

Suppose equality held for every such pair. Then, for each fixed q , every increment of length q would have the same value: for $0 < x < 1 - q$, take $p = 1 - q - x$, and extend to $x = 0$ by continuity. Thus

$$h(x + q) - h(x) = h(q) - h(0) = h(q) \quad (x, q \geq 0, x + q \leq 1).$$

Hence h is additive on $[0, 1]$. Continuity and $h(1) = 1$ force $h(t) = t$, contrary to nonaffinity. Therefore strict inequality occurs for some p, q . $\square$

We need two elementary finite-space facts.

Lemma 6.2 (Strict monotonicity on a finite space). *Let h be strictly increasing. On a finite probability space, if $Z \geq W$ and $\mathbb{P}(Z > W) > 0$, then $\mathbb{E}_h[Z] > \mathbb{E}_h[W]$.*

Proof. For every t , $\{W > t\} \subseteq \{Z > t\}$. If $Z(\omega) > W(\omega)$ on a state of positive mass, then on the nonempty interval $(W(\omega), Z(\omega))$ the two tail probabilities differ strictly. Since h is strictly increasing, the difference of the two integrands in (1) is positive on a set of positive Lebesgue measure. $\square$

Lemma 6.3 (Attainment of the finite dual). *Let I, J be finite, let $c = (c_{ij})_{I \times J}$ be a finite real matrix, and let the marginal laws on I and J be fixed. For every distortion h , the finite dual*

$$\sup_{u_i + v_j \leq c_{ij}} \{\mathbb{E}_h[u(I)] + \mathbb{E}_h[v(J)]\}$$

is attained.

Proof. The objective is invariant under $(u, v) \mapsto (u + a, v - a)$. Given a feasible pair, replace it successively by

$$\tilde{u}_i = \min_j (c_{ij} - v_j), \quad \tilde{v}_j = \min_i (c_{ij} - \tilde{u}_i).$$

Each replacement raises the corresponding potential coordinatewise and preserves feasibility, so the objective does not decrease. Fix $i_0 \in I$ and normalize $\tilde{u}_{i_0} = 0$. If $|c_{ij}| \leq M$, then

$$|\tilde{u}_i - \tilde{u}_{i_0}| \leq \max_j |c_{ij} - c_{i_0j}| \leq 2M,$$

and the identity $\tilde{v}_j = \min_i (c_{ij} - \tilde{u}_i)$ then gives $|\tilde{v}_j| \leq 3M$. Thus the supremum may be restricted to a compact set. Finally, distorted expectation is 1-Lipschitz in the sup norm by monotonicity and cash invariance, so the objective is continuous. $\square$

Theorem 6.4 (Necessity of submodularity). *Let h be continuous, strictly increasing, convex, and nonaffine. Let $c : \mathbb{R}^2 \rightarrow \mathbb{R}$ be bounded, continuous, and componentwise nondecreasing. If*

$$P_h(c; \mu, \nu) = D_h(c; \mu, \nu) \quad (22)$$

holds for every pair of two-point marginals μ, ν , then c is submodular. The same conclusion holds for componentwise nonincreasing c .

Proof. Suppose submodularity fails. Then there are $x_0 < x_1$ and $y_0 < y_1$ such that, with $c_{ij} = c(x_i, y_j)$,

$$c_{00} + c_{11} > c_{01} + c_{10}. \quad (23)$$

Monotonicity and (23) imply

$$A := c_{11} - c_{01} > 0, \quad B := c_{11} - c_{10} > 0. \quad (24)$$

Choose $p, q > 0$, $p + q < 1$, for which (21) is strict, and set

$$\mu = p\delta_{x_0} + (1-p)\delta_{x_1}, \quad \nu = q\delta_{y_0} + (1-q)\delta_{y_1}.$$

Every coupling is parametrized by $t \in [0, \min(p, q)]$:

$$\pi_{00} = t, \quad \pi_{01} = p - t, \quad \pi_{10} = q - t, \quad \pi_{11} = 1 - p - q + t. \quad (25)$$

Because h is continuous, the distorted expectation of the four-valued cost is continuous in t , so the primal minimum is attained.

Consider the relaxed finite dual in which u_i and v_j are specified only on the two support points. Restricting global feasible potentials gives

$$D_h(c; \mu, \nu) \leq D_h^{\text{fin}} \leq P_h(c; \mu, \nu),$$

where the second inequality is finite-space weak duality. Under the assumed global equality, all three quantities coincide. By Lemma 6.3, choose an optimal finite pair (u, v) , and choose an optimal coupling parameter t . Let $U = u_i$, $V = v_j$, and $C = c_{ij}$ on cell (i, j) . Equality of the optimal values forces equality throughout

$$\mathbb{E}_h[C] \geq \mathbb{E}_h[U + V] \geq \mathbb{E}_h[U] + \mathbb{E}_h[V]. \quad (26)$$

By Lemma 6.2, the first equality implies contact on every positive-mass cell:

$$u_i + v_j = c_{ij} \quad \text{whenever } \pi_{ij} > 0. \quad (27)$$

If $t > 0$, then cells $(0, 0)$ and $(1, 1)$ have positive mass. Equality there, together with feasibility on the two off-diagonal cells, gives

$$c_{00} + c_{11} = (u_0 + v_0) + (u_1 + v_1) \leq c_{01} + c_{10},$$

contradicting (23). Hence $t = 0$. The remaining three cells $(0, 1)$, $(1, 0)$, $(1, 1)$ have positive mass, so contact gives

$$u_1 - u_0 = A, \quad v_1 - v_0 = B.$$

Let $I = \mathbf{1}_{\{X=x_1\}}$ and $J = \mathbf{1}_{\{Y=y_1\}}$. Then $U = u_0 + AI$, $V = v_0 + BJ$, and

$$\mathbb{P}(I = 1) = 1 - p, \quad \mathbb{P}(J = 1) = 1 - q, \quad \mathbb{P}(I = J = 1) = 1 - p - q.$$

If $A \leq B$, a direct three-level Choquet calculation gives

$$\begin{aligned}\mathbb{E}_h[U + V] &= u_0 + v_0 + A + (B - A)h(1 - q) + Ah(1 - p - q), \\ \mathbb{E}_h[U] + \mathbb{E}_h[V] &= u_0 + v_0 + Ah(1 - p) + Bh(1 - q).\end{aligned}$$

Their difference is $A\Delta_h(p, q) > 0$. If $B \leq A$, the symmetric calculation gives $B\Delta_h(p, q) > 0$. Equivalently,

$$\mathbb{E}_h[U + V] - \mathbb{E}_h[U] - \mathbb{E}_h[V] = \min\{A, B\}\Delta_h(p, q) > 0, \quad (28)$$

contradicting the second equality in (26). Therefore no violating rectangle exists, and c is submodular.

If c is componentwise nonincreasing, apply the preceding argument to $(x, y) \mapsto c(-x, -y)$. $\square$

Corollary 6.5 (Sharp characterization in the monotone class). *Let h be continuous, strictly increasing, convex, and nonaffine. For a bounded continuous componentwise-monotone cost c , the identity*

$$P_h(c; \mu, \nu) = D_h(c; \mu, \nu) \quad \text{for every } \mu, \nu \in \mathcal{P}(\mathbb{R})$$

holds if and only if c is submodular.

Proof. Sufficiency is Theorem 5.4; necessity is Theorem 6.4. $\square$

Remark 6.6. *The strict-increase assumption in Corollary 6.5 is essential for this conclusion. For $0 \leq \lambda < 1$, the universal distortions h_λ have a flat segment, and Theorem 4.1 shows that no submodularity assumption is needed at all.*

7 Pullbacks and nonmonotone examples

The monotonicity needed in Theorem 5.4 may hold only after passing to sufficient statistics.

Corollary 7.1 (Pullback class). *Let h be convex, let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be Borel, and let $\bar{c} : \mathbb{R}^2 \rightarrow \mathbb{R}$ be bounded, continuous, componentwise monotone, and submodular. Set*

$$c(x, y) = \bar{c}(f(x), g(y)).$$

Then strong duality holds for c and every pair of marginals.

Proof. Let $\alpha = f_{\#}\mu$ and $\beta = g_{\#}\nu$. Lemma 2.2 shows that every coupling of $(f(X), g(Y))$ can be lifted to a coupling of (X, Y) . Therefore the primal for c and (μ, ν) equals the primal for $\bar{c}$ and (α, β) . Apply Theorem 5.4. Composing the resulting dual potentials with f and g gives bounded Borel feasible potentials for c with the same marginal distorted expectations. Weak duality completes the proof. $\square$

For example,

$$c(x, y) = 1 - \exp\{-e^{\sin x} - e^{\cos y}\}$$

is neither componentwise monotone nor separable in (x, y) , but it belongs to the pullback class and therefore satisfies strong duality for every convex distortion.

8 Conclusion and remaining scope

The affine restriction in the previously known strong-duality theorem is not fundamental. The arguments above establish the following.

1. Every bounded separable cost satisfies strong duality for every convex distortion.
2. The family h_λ , $0 \leq \lambda \leq 1$, is exactly the family of distortions for which every bounded continuous cost satisfies the given duality identity for all marginals.
3. For an arbitrary convex distortion, every bounded continuous componentwise-monotone submodular cost satisfies strong duality and has bounded monotone dual optimizers.
4. For continuous strictly increasing convex nonaffine distortions, submodularity is necessary and sufficient within the bounded continuous componentwise-monotone class.

These statements constitute substantial partial progress on the question of Liu, Wang, Wang, and Zhuang, but not a complete solution. In particular, they do not characterize all pairs (h, c) . The unresolved region includes general nonmonotone, nonseparable costs for a fixed strictly increasing convex distortion, as well as broader measurable or unbounded settings and general nonconvex distortions.

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